

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding must select the event occurrence answering a query, even when opposite event orders co-exist in one recording. Returning the same window for inverse queries reveals a query-window binding error. We introduce RelTwin: shared excerpts form opposite-order windows whose timestamp answers compete under each query. Both windows have acoustic support; the requested order determines the correct answer. Candidate supervision complements sequence learning while retaining direct timestamp generation. Against paired SFT, RelTwin improves joint localization by 31.25 points on a synthetic relation-development set and SpotSound mIoU by 1.02 points, with binding gains retained under timing intervention. Complementary system evaluations of NOVA with selective refinement reach 59.39, 86.86, and 73.46 mIoU on SpotSound, Clotho-Moment, and the released UnAV subset, respectively. Controlled training comparisons identify the contribution of query-conditioned answer competition.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query-window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query-window binding error*. We target this distinction between event detection and query-specific localization.

RelTwin (Fig. 1) addresses this binding problem through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Each window contains a real event sequence, but its correctness as an answer depends on the query. Sequence supervision rewards correct timestamp answers individually; the added candidate loss explicitly compares the two answers under each query.

This local competition trains query-window binding while inference retains direct timestamp generation.

Our contributions are the characterization of query-window binding errors, locally valid timestamp supervision, and matched training and timing evidence of binding gains. NOVA evaluations on SpotSound, Clotho-Moment, and UnAV complement this mechanism study with system-level grounding results.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions [3]. Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned generation. TimeAudio adds temporal markers and absolute time-aware encoding [7]. SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8], combining presence supervision with timestamp prediction. RelTwin targets the query-specific choice between occurrences when inverse relations both have acoustic support.

Compositional audio-language learning. CLAP aligns audio and text representations [9]; T-CLAP adds order-reversed descriptions and temporal-focused contrast [10]. CompA evaluates event order and attribute binding, while CompA-CLAP trains with composition-aware negatives and modular contrast [11]. CoSTALA combines spatio-temporal contrast, local event-caption alignment, and feature consistency [12]. These methods organize compositional audio-text representations. RelTwin instead compares the query-conditioned likelihoods of complete timestamp answers, supervising selection in a grounding model’s output space.

Hard negatives for temporal grounding. For compositional video grounding, SHINE combines generated hard-negative queries with within-video and cross-query saliency ranking [13], comparing segment relevance and query compositions. RelTwin’s candidates are complete timestamp answers within one recording. Each denotes a real event sequence; reversing the query exchanges their labels while preserving the audio. Sequence likelihoods train this query-specific choice.

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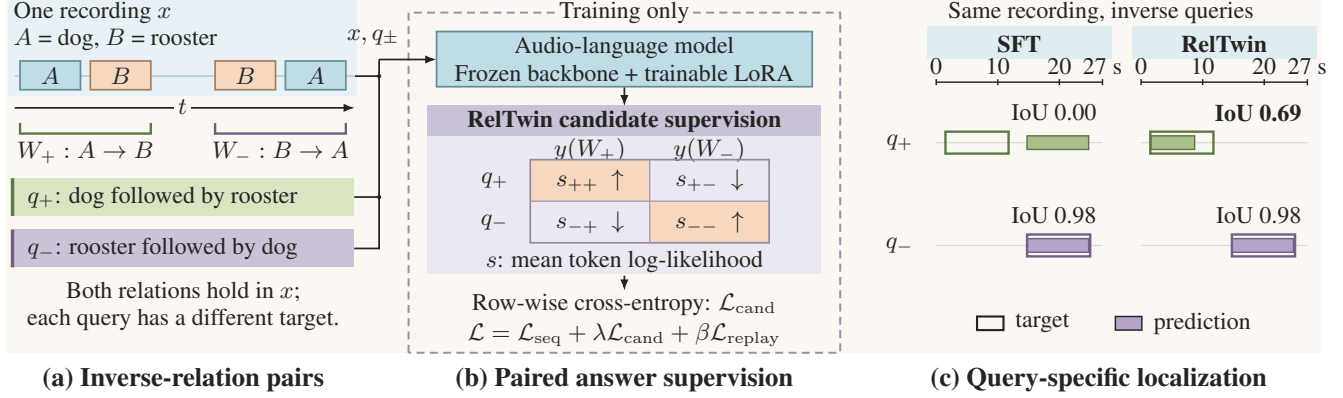


Fig. 1. RelTwin binds inverse queries to different windows in one recording. (a) Reused excerpts form two local targets. (b) The dashed enclosure marks training-only computation; shaded diagonal cells identify correct timestamp answers, and up/down arrows indicate training preferences. (c) In a relation-development example, SFT repeats one answer while RelTwin selects the corresponding window for each query. Outlines mark targets; filled bars show predictions, evaluated by set-IoU.

3. QUERY-SPECIFIC WINDOW LEARNING

3.1. Two true relations, two different answers

In one recording x , RelTwin arranges the same ESC-50 excerpts [14], A and B , as $A \rightarrow B$ and $B \rightarrow A$. Their local targets W_+ and W_- run from the first excerpt’s onset to the second’s offset, including the gap. Queries q_+ and q_- request the respective orders. Both windows have acoustic support, but W_- is positive for q_- and negative for q_+ ; reversing the query exchanges their labels.

The templates “followed by” and “and then” yield two inverse-query pairs per recording. Earlier-order layouts alternate (Section 5.2). Category rehearsal targets the union of both occurrences of A or B , retaining category-grounding supervision.

3.2. Contrast timestamp answers during training

We serialize an interval set S as comma-separated spans, each formatted like from 1.500s to 11.750s (three-decimal precision). Its supervised assistant-completion tokens $y(S)$ exclude the prompt and are scored by

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix M , with entries $M_{ij} = s_\theta(x, q_i, W_j)$ for $i, j \in \{+, -\}$, indexes queries by rows and answers by columns. Its diagonal labels $Y = I_2$ identify the correct query-answer pairs (Fig. 1b). Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$, rewarding the two correct responses individually. It does not explicitly compare them with the other locally valid window. RelTwin trains the relative preferences $M_{++} > M_{+-}$ and $M_{--} > M_{-+}$ through row-wise probabilities $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$, where

$\tau > 0$. The candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

For a separately sampled category example (x_r, q_r, S_r) , the rehearsal loss is $\mathcal{L}_{\text{replay}} = -s_\theta(x_r, q_r, S_r)$. Together,

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Paired SFT sets $\lambda = 0$; RelTwin uses $\lambda > 0$ with identical relation and rehearsal data. Both directly generate timestamps at inference, without candidates or an additional scorer.

3.3. Joint localization and relation selection

For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, define $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. We report pair-level percentages; same-answer counts use exact equality of parsed interval lists.

Predicting the union of two disjoint, equal-length targets for both queries passes PairAcc (IoU 0.5) but fails JointPairAcc. Inspired by paired compositional evaluation [15], JointPairAcc distinguishes query-specific selection from coverage.

4. EXPERIMENTAL SETUP

Data and evaluation protocol. Adaptation uses 40 ESC-50 classes: 256 recordings, 1,024 relation queries (512 pairs), and 512 rehearsal queries. Ten held-out classes provide 80 recordings, 320 queries, and 160 relation-development pairs, with disjoint original Freesound IDs. System evaluation uses 400 SpotSound queries, 6,649 Clotho-Moment queries, and

Table 1. System-level grounding (%; higher is better). Baselines: SpotSound Table 3 [8]; NOVA configurations: Section 4.1; controlled RelTwin study: Table 2. Bold marks column maxima. †UnAV is the released 100-query subset, with protocol-qualified published-score comparisons (Section 4). Deltas use per-metric prior main-table bests.

Method	SpotSound-Bench			Clotho-Moment			UnAV subset [†]		
	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5
<i>Task-specific grounding models</i>									
WTATG	38.4	53.0	34.5	9.1	12.1	6.3	38.4	53.0	37.0
AM-DETR	19.5	25.5	15.0	80.9	89.8	88.0	42.8	59.0	46.0
<i>Proprietary audio–language models</i>									
Gemini-2.5-Flash	25.7	34.8	27.8	36.9	45.7	33.7	35.6	48.0	37.0
Gemini-2.5-Pro	19.7	20.0	17.3	32.5	40.4	32.5	34.6	39.0	35.0
<i>General-purpose open audio–language models</i>									
Kimi-Audio	3.2	3.0	1.3	0.9	0.7	0.1	5.3	8.0	2.0
Qwen2-Audio	4.4	5.5	1.8	5.7	6.1	1.3	9.7	14.0	4.0
Audio Flamingo 3	7.6	8.8	2.8	22.6	32.9	21.8	25.0	35.0	24.0
<i>Temporal grounding-enhanced models</i>									
TimeAudio	10.2	7.8	1.8	28.6	39.5	24.9	16.0	21.0	7.0
SpotSound-Q	49.9	65.0	51.3	85.4	93.6	91.2	72.4	88.0	74.0
SpotSound-A	57.9	76.5	59.5	85.6	93.4	91.0	69.8	86.0	74.0
NOVA + selective refinement	59.39	79.25	62.50	86.86	93.49	90.84	73.46	89.00	80.00
Prior main-table best	57.9	76.5	59.5	85.6	93.6	91.2	72.4	88.0	74.0
Δ vs. prior best	+1.49	+2.75	+3.00	+1.26	−0.11	−0.36	+1.06	+1.00	+6.00

the released 100-query/77-audio UnAV subset [2]. UnAV scores clip intervals to audio duration (unclipped mIoU: 73.39). SpotSound lists 997 UnAV queries; exact sample equivalence is unconfirmed without its manifest. Development and public results informed project iterations.

Training settings. SFT and RelTwin initialize from SpotSound-A on Audio Flamingo 3. LoRA [16] uses rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Both run 256 AdamW [17] updates at learning rate 5×10^{-6} , zero weight decay, gradient clipping 1.0, $\tau = 1$, $\beta = 0.5$, and RelTwin $\lambda = 1$. Each update accumulates one inverse-query pair and one rehearsal query, not multiple pairs. Seed-0 comparisons match initialization, data order, code, environment, and updates, not FLOPs or wall time: RelTwin scores four combinations without gradients then recomputes weighted gradients; SFT backpropagates two positive answers. Both include rehearsal.

Inference. SFT seed 0 and RelTwin seeds 0/1/2 are evaluated after 256 updates. All generators use 16 kHz mono audio, bfloat16, and greedy one-beam decoding without sampling, capped at 128 new tokens.

Comparison roles. Table 1 reports system performance; Table 2 isolates candidate supervision. The latter backbone scores 58.42 mIoU versus published 57.9; RelTwin’s three-seed difference against it is +0.75 points (audio-group CI $[-0.24, 1.74]$).

Statistical analysis. Training comparisons use 20,000 audio-group bootstrap resamples for 95% CIs. Relation comparisons resample 39 connected source-sharing groups

(largest: 18 audios), preserving reused excerpts. CIs quantify sample uncertainty for evaluated seeds, without development-selection correction.

4.1. NOVA system configurations

NOVA (Necessity-Oriented Verification over Audio) combines candidate generation, verification, and refinement. SpotSound uses a RelTwin-data training extension with Jensen–Shannon consistency between p_+ and label-swapped p_- added to Eq. 3. Setwise preference optimization (SetPO) continues for 64 updates at 2×10^{-6} over six candidates: both windows, their expansions, enclosing span, and union. Soft targets combine set-IoU and interval-coverage quality. Table 2 tests Eq. 3 without these extensions.

Official, SFT, and SetPO seed-1 SpotSound predictions enter a LongNeedle-ESC50-fitted ridge router, using geometry and frozen query-existence scores after keeping/removing candidate intervals. Clotho and UnAV use official predictions without this router. Refinement uses frozen SetPO/SpotSound features and ridge edit utilities: dynamic programming selects compatible boundaries, accepted when the 200-model bootstrap 5th percentile exceeds 0.02. Event count is preserved; radius $\min(0.25 \text{ s}, 0.625 \text{ event duration})$ is shared. Audio-Grounding/AEGBench development evaluations selected the duration coefficient before these applications. Table 1 is an extra-data system comparison.

Table 2. Query–window binding on the synthetic relation-development set and transfer to SpotSound. Panel (b) compares seed-0 models with matched data and updates. Panel (c) reports means $\pm$ sample SD; same-answer counts follow seed order 0/1/2. Joint denotes JointPairAcc; same-answer counts identical timestamp answers to inverse queries. Scores are percentages except pair counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison: seed 0</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
RelTwin	59.43	88.55	84.38	7.50	1/160
RelTwin – Paired SFT	+1.02	+13.97	+31.25	–32.50	–36 pairs
<i>(c) RelTwin replication: three training seeds</i>					
RelTwin	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

5. RESULTS AND ANALYSIS

5.1. Timestamp contrasts recover query-specific selection

Paired SFT returns identical answers for 37/160 inverse-query pairs across 22 recordings; RelTwin reduces this to one (Table 2). Of those 37 pairs, 27 become jointly correct: differentiation recovers the requested relation, not just different timestamps. Figure 1 illustrates this recovery; residual under-coverage distinguishes correct relation selection from exact boundaries.

JointPairAcc increases by 31.25 points (source-group CI [22.35, 40.50]) and Swap error decreases by 32.50. Across all pairs, 51 become jointly correct, one regresses, and 24 remain incorrect. Relation mIoU increases by 13.97 points (CI [9.84, 17.51]), connecting selection with overlap.

Across three RelTwin seeds, JointPairAcc averages 81.46 and identical-answer counts are 1, 0, and 0, reproducing the reduction in same-answer collapse.

5.2. Binding gains persist after timing intervention

To test layout-specific silence, we recompose 80 development audios with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving waveforms, order, and queries. Frozen seed-0 models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

On changed recordings, RelTwin reaches 87.50 JointPairAcc against SFT’s 67.50 (+20.00; CI [2.94, 33.33]); originally they scored 85.00 and 42.50. The gap narrows but remains positive, supporting binding beyond the original silence layout. Across all 160 queries, SFT and RelTwin score 81.34 and 90.65 mIoU. This diagnostic follows method development.

5.3. System performance and training transfer

NOVA reaches 59.39 SpotSound mIoU (Table 1), exceeding the prior main-table best by 1.49 points; R1@.3/.5 improve by 2.75/3.00. Clotho reaches 86.86 mIoU (+1.26), while recalls trail prior bests. UnAV reaches 73.46 (+1.06), subject to the release qualification in Section 4.

RelTwin’s matched comparison supplies training attribution: +1.02 SpotSound mIoU over SFT (CI [0.23, 1.81]), with 92 wins, 239 ties, and 69 losses, including two drops $\geq$ 0.5 IoU. Three-seed mIoU is 59.16, exceeding published 57.9 by 1.26. Clotho/UnAV instead start at 86.8543/73.4513 with official predictions, gaining 0.0013/0.0133 from refinement. These system totals and controlled training gains answer different questions: deployment performance and candidate supervision’s effect.

5.4. Scope

Repeated excerpts and fixed templates enable controlled binding tests. Alternative hard-negative constructions, matched multi-seed replication, and natural overlapping or long-range relations remain open evaluations. Adaptation-stage source separation does not audit backbone pretraining overlap.

6. CONCLUSION

RelTwin learns query–window binding by contrasting locally valid timestamp answers. Matched SFT and timing comparisons show improved relation selection and SpotSound transfer; NOVA evaluations provide complementary system results. Explicit answer competition connects acoustic support to query-specific localization.

7. ACKNOWLEDGMENTS

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Compliance with Ethical Standards. We reuse public ESC-50 [14], SpotSound [8], Clotho-Moment and UnAV [2], AudioGrounding [1], and AEGBench [18] data, including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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